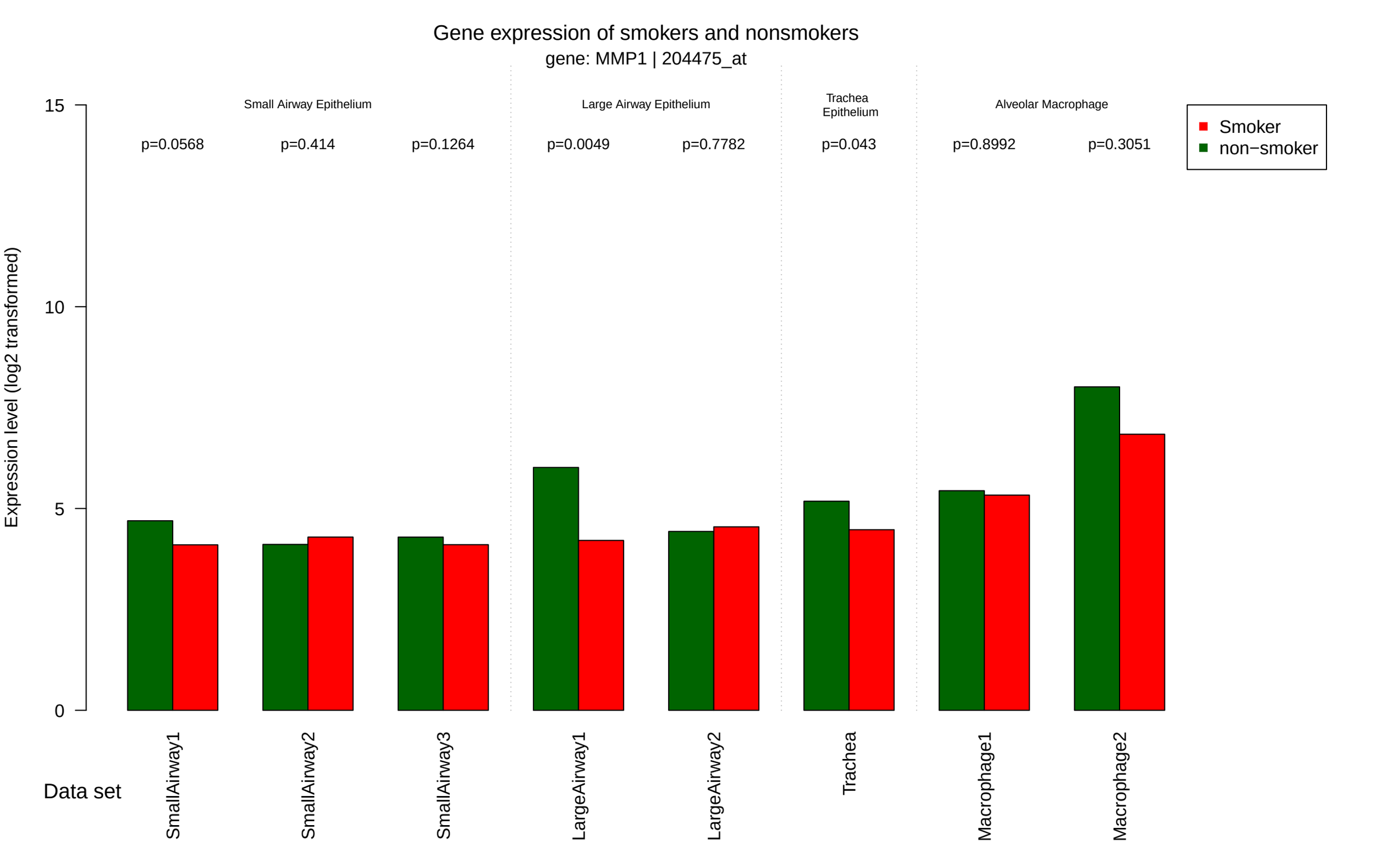
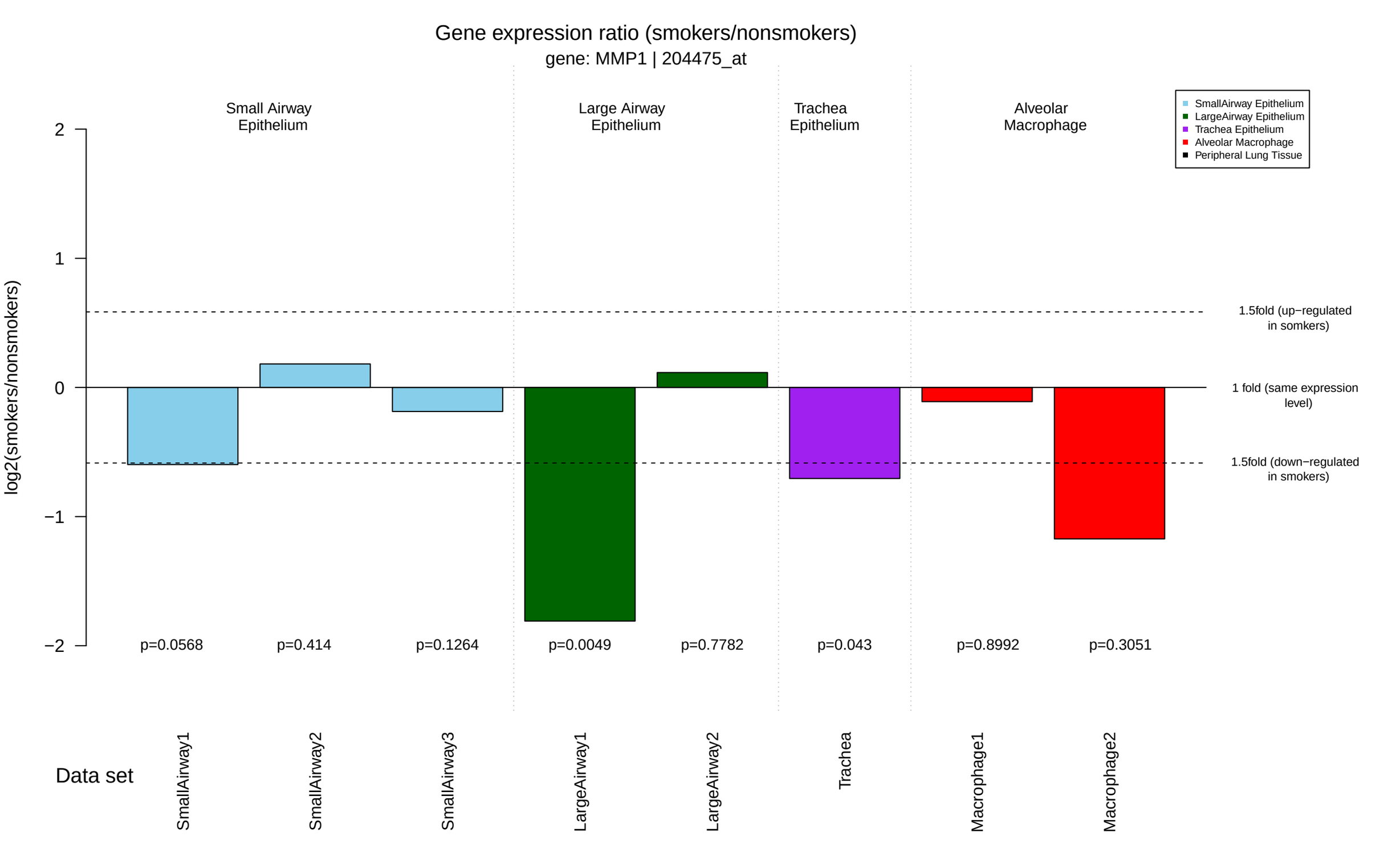
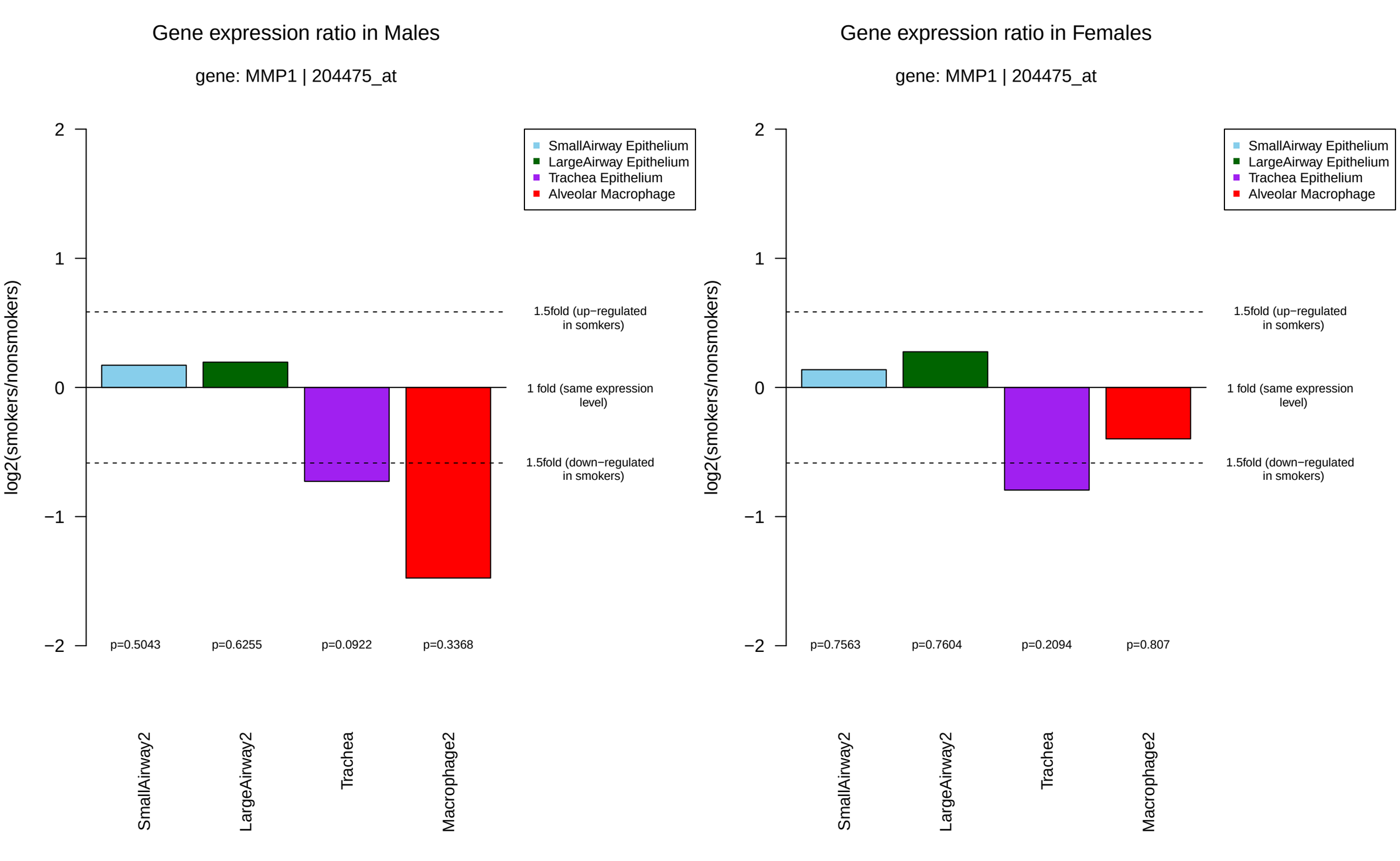
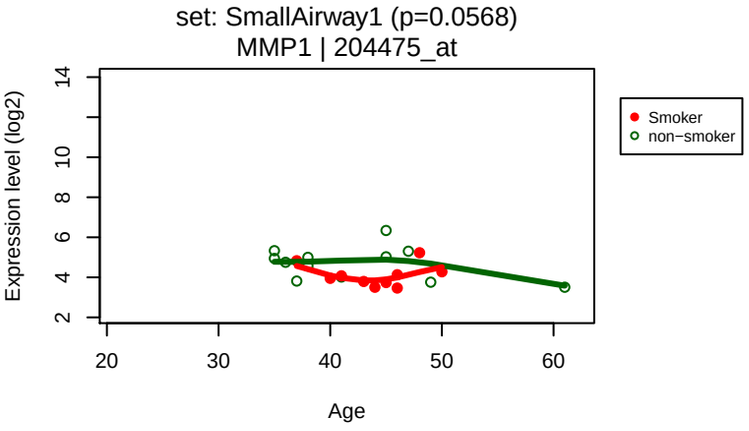
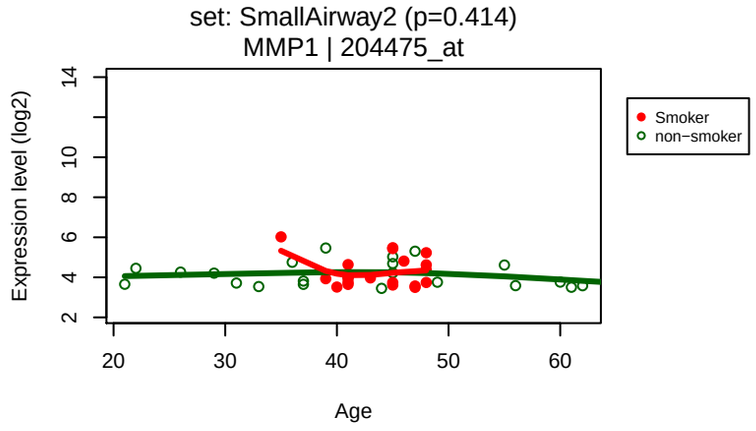
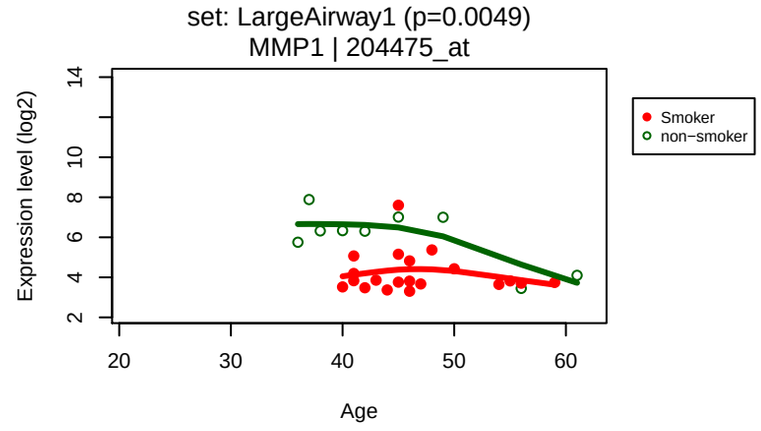

Gene expression of smokers and nonsmokers
gene: MMP1 | 204475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0568
p=0.414
p=0.1264
p=0.0049
p=0.7782
p=0.043
p=0.8992
p=0.3051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

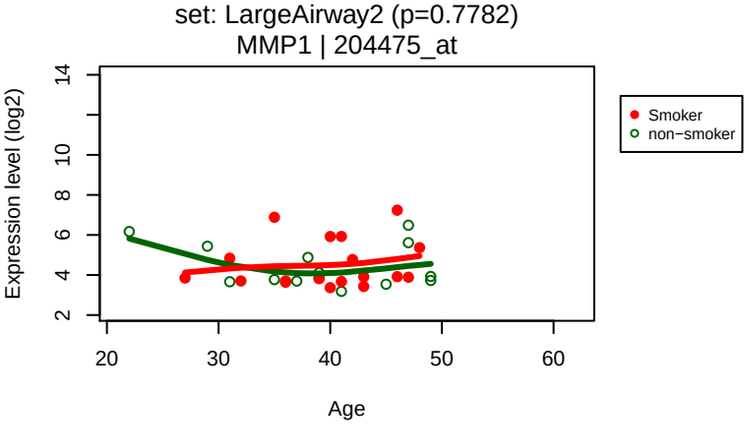

Gene expression ratio (smokers/nonsmokers)
gene: MMP1 | 204475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0568
p=0.414
p=0.1264
p=0.0049
p=0.7782
p=0.043
p=0.8992
p=0.3051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

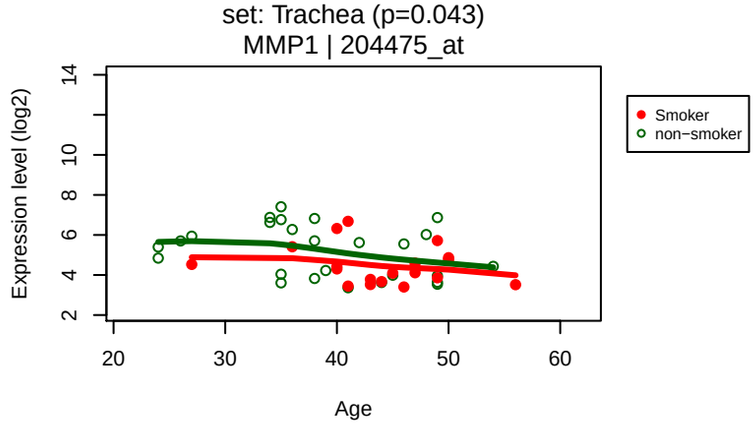

Gene expression ratio in Males
gene: MMP1 | 204475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5043
p=0.6255
p=0.0922
p=0.3368
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP1 | 204475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7563
p=0.7604
p=0.2094
p=0.807
SmallAirway2
LargeAirway2
Trachea
Macrophage2

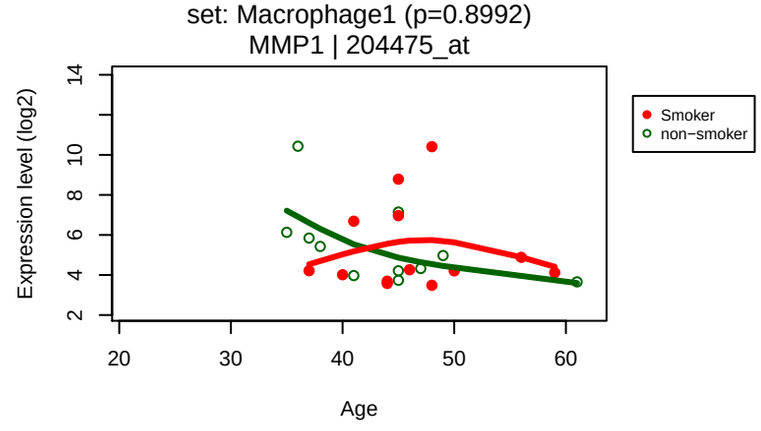

set: SmallAirway1 (p=0.0568)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker

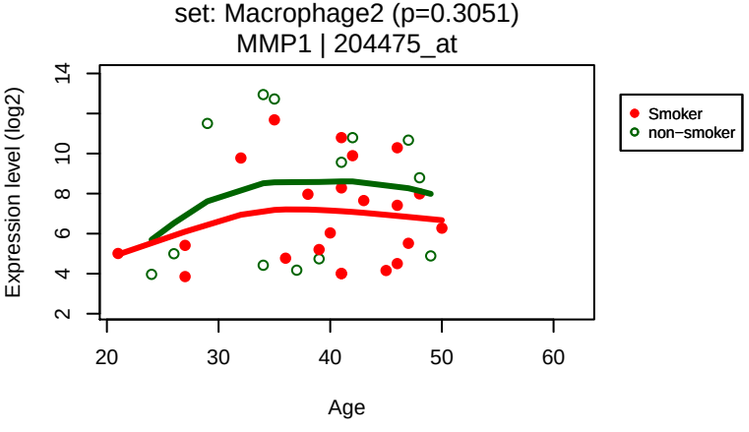

set: SmallAirway2 (p=0.414)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker

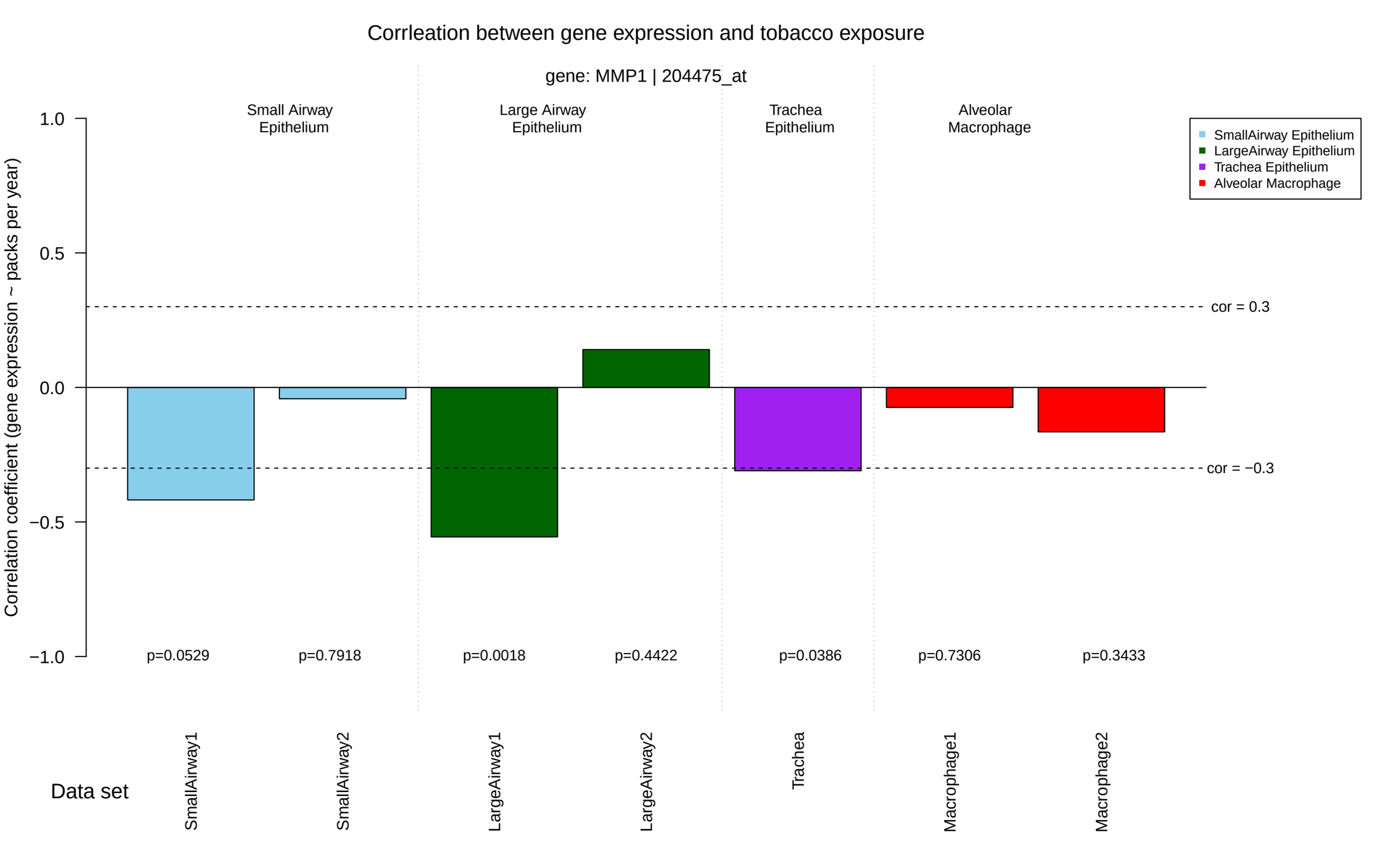

set: LargeAirway1 (p=0.0049)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7782)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.043)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8992)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3051)
MMP1 | 204475_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP1 | 204475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0529
p=0.7918
p=0.0018
p=0.4422
p=0.0386
p=0.7306
p=0.3433
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2